No.

Date:

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Kelas : 1A TNK

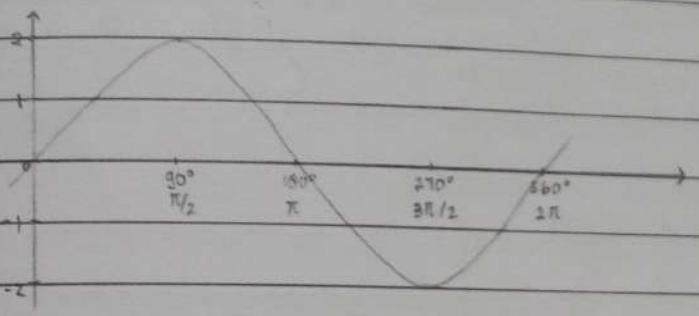
Latihan 3

Bagian 1. Tentukan amplitudo, periode dan gambarkan grafik fungsi trigonometri (manual dan bantuan alat) berikut untuk satu periode !

1. $y = 2 \sin x$

amplitudo = 2

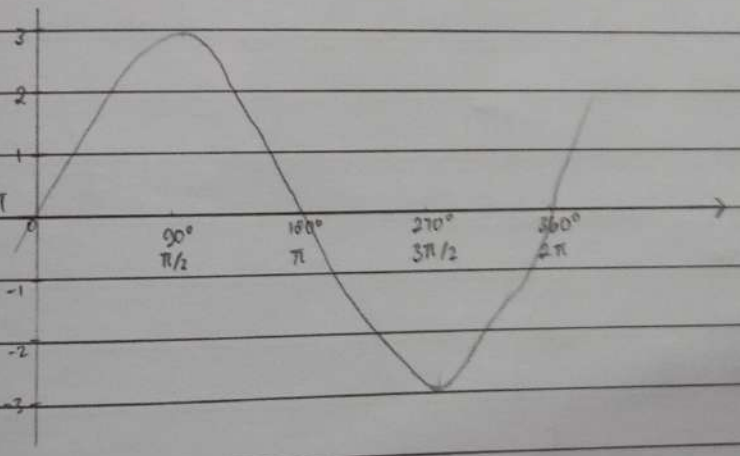
periode $= \frac{2\pi}{1} = 2\pi$



2. $y = 3 \sin x$

Amplitudo = 3

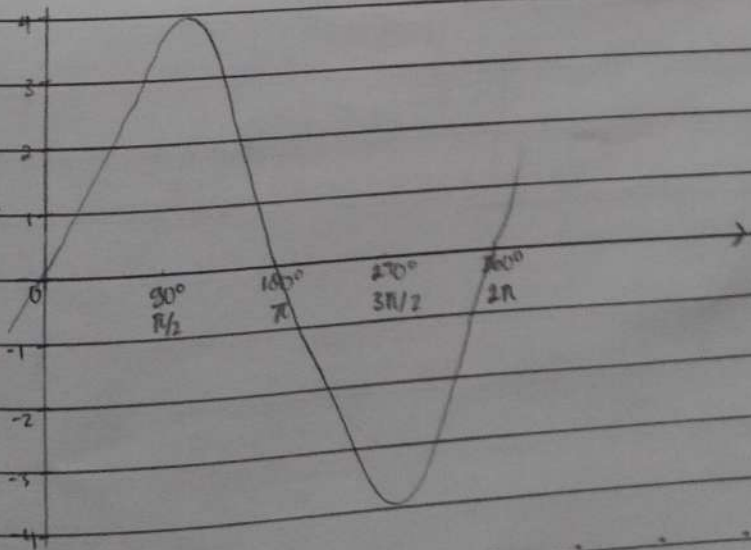
periode $= \frac{2\pi}{1} = 2\pi$



3. $y = 4 \sin x$

Amplitudo = 4

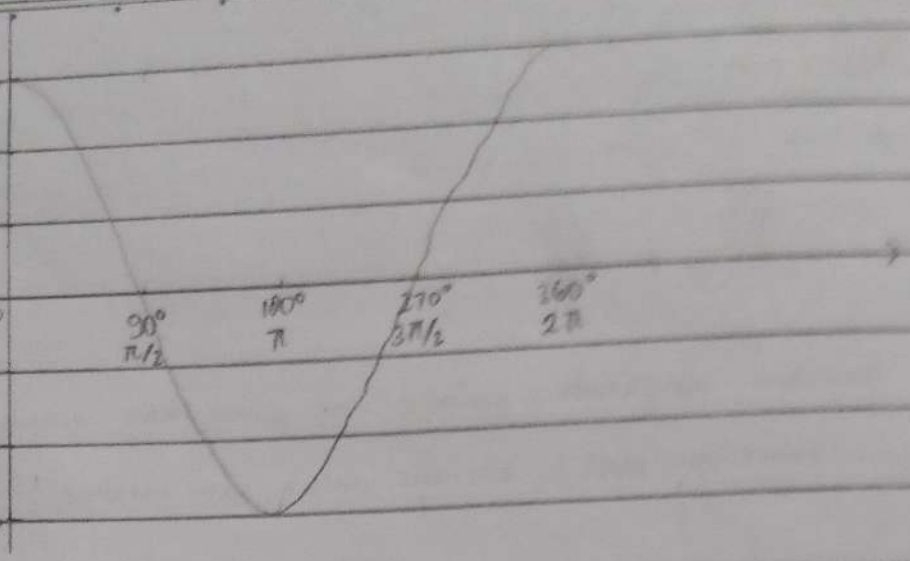
periode $= \frac{2\pi}{1} = 2\pi$



4. $y = 6 \cos x$

Amplitude = 6

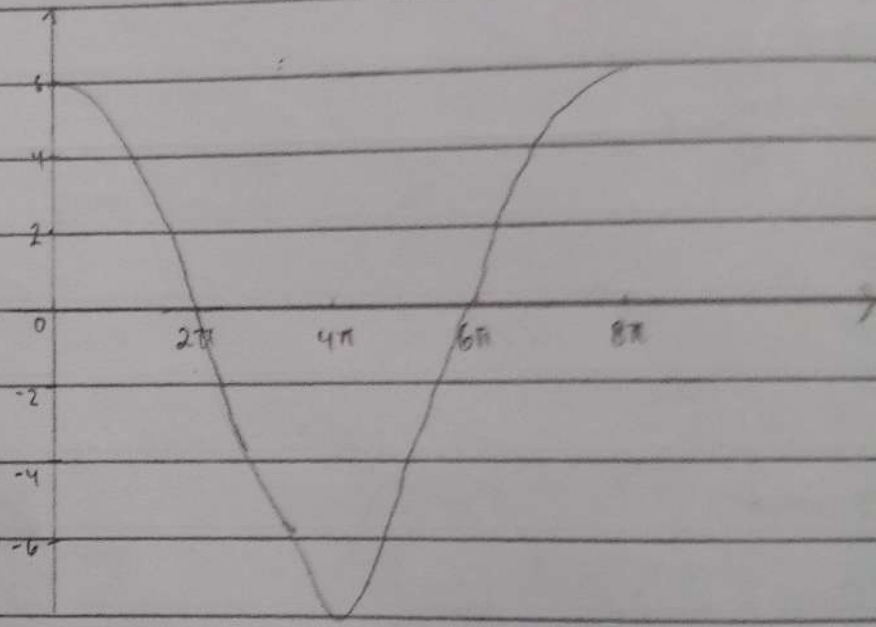
period = $\frac{2\pi}{1} = 2\pi$



5. $y = 6 \cos \frac{x}{4}$

Amplitude = 6

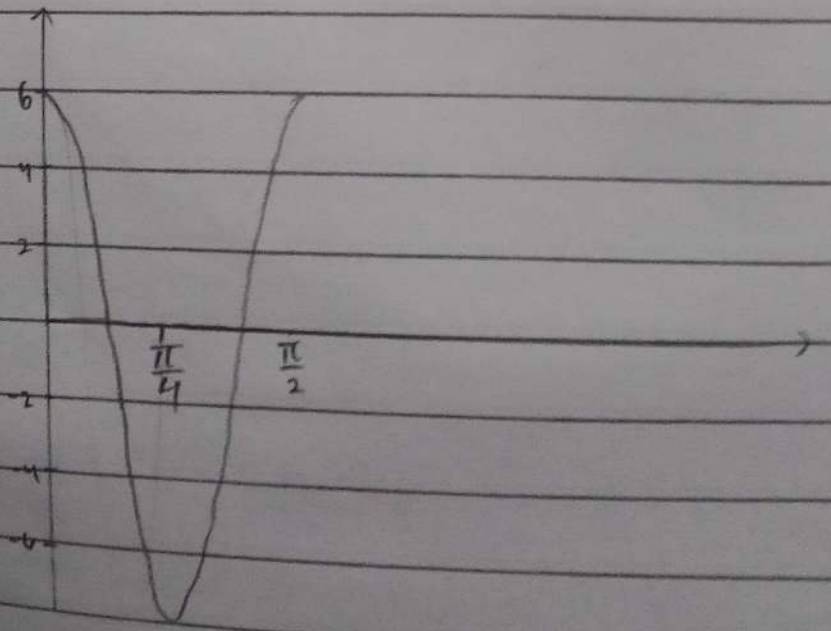
period = $\frac{2\pi}{\frac{1}{4}} = 2\pi \cdot 4 = 8\pi$



6. $y = 6 \cos 4x$

Amplitude = 6

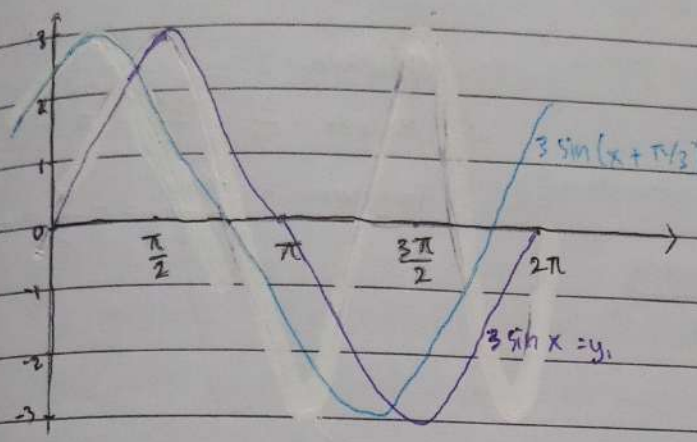
period = $\frac{2\pi}{4} = \frac{\pi}{2}$



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Bagian II. Gunakan prinsip pergeseran dari bentuk $y_1 = R \sin x$ atau $y_2 = R \cos x$ untuk grafik fungsi trigonometri untuk satu periode berikut :

1. $y = 3 \sin (x + \pi/3)$

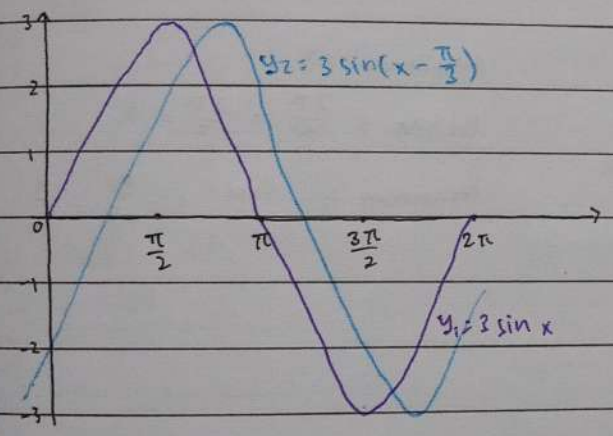


Amplitudo : 3

Periode : 2π

Pergeseran : bergeser ke kiri sebanyak $\frac{\pi}{3}$

2. $y = 3 \sin (x - \frac{\pi}{3})$

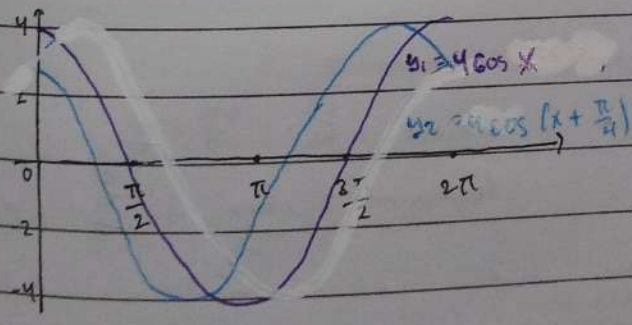


Amplitudo = 3

Periode = 2π

Pergeseran : bergeser ke kanan sebanyak $\frac{\pi}{3}$

3. $y = 4 \cos (x + \frac{\pi}{4})$

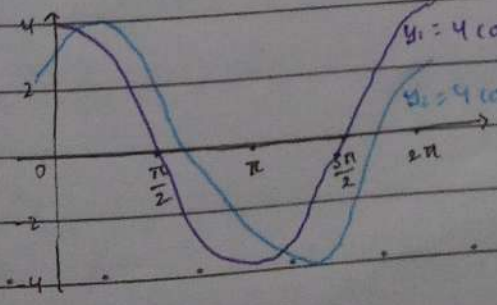


Amplitudo = 4

Periode = 2π

Pergeseran : bergeser ke kiri sebanyak $\frac{\pi}{4}$

4. $y = 4 \cos (x - \frac{\pi}{4})$



Amplitudo = 4

Periode = 2π

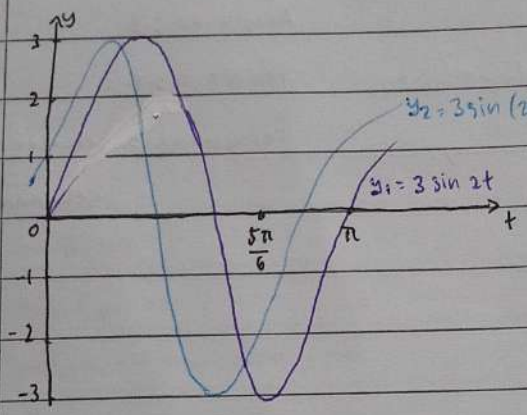
Pergeseran : bergeser ke kanan sebanyak $\frac{\pi}{4}$

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Bagian II Gunakan prinsip pergeseran dari bentuk $y_1 = R \sin \omega t$ atau $y_2 = R \cos \omega t$ untuk grafik fungsi trigonometri untuk satu periode, dan berapa amplitudo dan periode (dalam detik) dengan ω (kecepatan sudut) serta lengkapi grafiknya berikut

1. $y = 3 \sin \left(2t + \frac{\pi}{3}\right)$

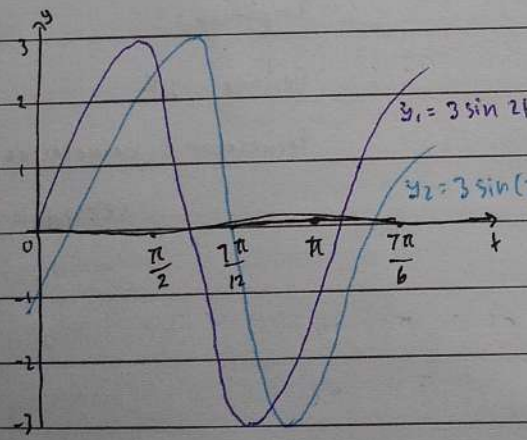


Amplitudo = 3

Periode $= \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$

Pergeseran $= -\frac{\text{Fase}}{\omega} = -\frac{\frac{\pi}{3}}{2} = -\frac{\pi}{6}$

2. $y = 3 \sin \left(2t - \frac{\pi}{3}\right)$

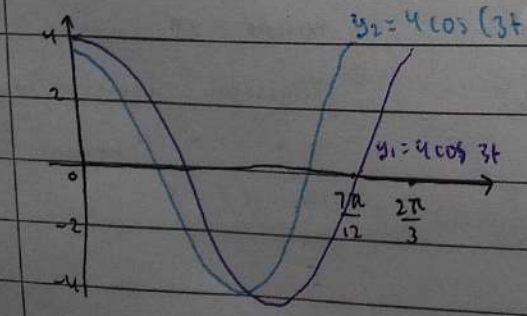


Amplitudo = 3

Periode $= \frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$

Pergeseran $= -\frac{\text{Fase}}{\omega} = -\frac{-\frac{\pi}{3}}{2} = \frac{\pi}{6}$

3. $y = 4 \cos \left(3t + \frac{\pi}{4}\right)$



Amplitudo = 4

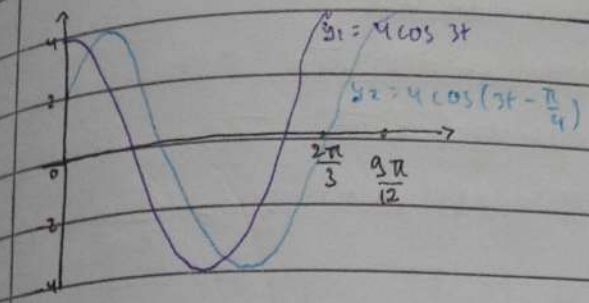
Periode $= \frac{2\pi}{\omega} = \frac{2\pi}{3} = \frac{2\pi}{3}$

Pergeseran $= -\frac{\text{Fase}}{\omega} = \frac{-\frac{\pi}{4}}{3} = -\frac{\pi}{12}$

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4. $y = 4 \cos (3t - \frac{\pi}{4})$



Amplitudo = 4

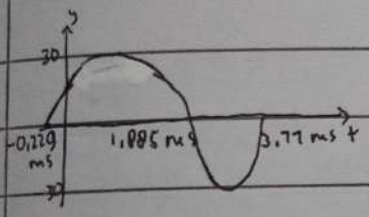
Periode = $\frac{2\pi}{\omega} = \frac{2\pi}{3}$

Pergeseran = $-\frac{\text{Fase}}{\omega} = -\frac{\frac{\pi}{4}}{3} = \frac{\pi}{3}$

Bagian III

Tentukan amplitudo, waktu periodik, frekuensi dan sudut phase (leading atau lagging R* sin ωt, dan grafik untuk fungsi-fungsi berikut :

1. $y = 30 \sin (50\pi t + 0{,}36)$



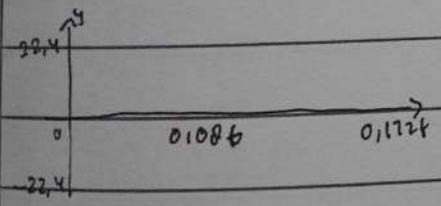
Amplitudo = 30

Waktu periodik = $\frac{2\pi}{\omega} = \frac{20\pi}{50\pi} = \frac{1}{25} = 0{,}04\,s = 4$ ms

Frekuensi = $\frac{1}{T} = \frac{1}{0{,}04} = 25$ Hz

Perbedaan fase = $-\frac{a}{\omega} = -\frac{0{,}36}{50\pi} = -0{,}00229$ (leading)

2. $x = 22{,}4 \sin (40t - 0{,}60)$



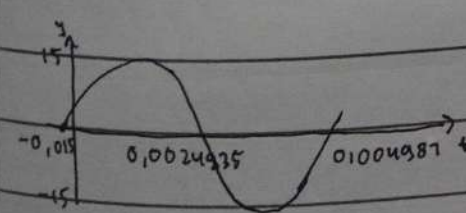
Amplitudo = 22,4

Waktu periodik = $\frac{2\pi}{40\pi} = \frac{1}{20\pi} = 0{,}05\,\pi$

Frekuensi = $\frac{1}{T} = \frac{1}{0{,}05\pi} = 20\pi$

Perbedaan fase = $-\frac{a}{\omega} = -\frac{0{,}6}{40} = 0{,}015$ (lagging)

3. $i = 15 \sin (314{,}2\,t + 0{,}468)$



Amplitudo = 15

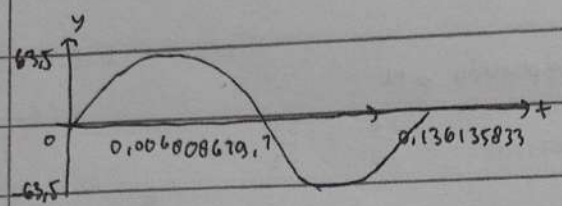
Waktu periodik = $\frac{2\pi}{314{,}2} = 0{,}019987$

Frekuensi = $\frac{1}{T} = \frac{1}{0{,}019987} = 50{,}03$

Perbedaan fase = $-\frac{a}{\omega} = -\frac{0{,}6}{40} = -0{,}015$ (leading)

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4. $y = 63,5 \sin (150\pi t - 0,132)$



Amplitudo = 63,5

Waktu periodik = $\frac{2\pi}{150\pi} = 0,01333$

Frekuensi = $\frac{1}{T} = \frac{1}{0,01333} = 75$

Perbedaan fase = $-\frac{a}{\omega} = \frac{0,132}{150\pi} = 0,000280325$ (lagging)

Bagian IV

1. Arus bolak balik rangkaian listrik memenuhi persamaan : $i(t) = 32 \sin(3\pi t)$ dengan $t \geq 0$

Hitung :

a. arus pada $t = 0,5$ ms dan $t = 2$ ms

$i(t) = 32 \sin(3\pi t)$

untuk $t = 0,5 \text{ ms} = 5 \cdot 10^{-4}$ s

$i(5 \cdot 10^{-4}) = 32 \sin(3\pi \cdot 5 \cdot 10^{-4})$

$= 32 \sin(15\pi \cdot 10^{-4})$

$= 32 \sin(0,0015\pi)$

$= 32 \cdot 8,2205 \cdot 10^{-5}$

$= 2,63056$ A

untuk $t = 2 \text{ ms} = 2 \cdot 10^{-3}$ s

$i(2 \cdot 10^{-3}) = 32 \sin(3\pi \cdot 2 \cdot 10^{-3})$

$= 32 \sin(6\pi \cdot 10^{-3})$

$= 32 \sin(0,006\pi)$

$= 32 \cdot 3,288 \cdot 10^{-4} = 0,01052$ A

b. waktu t pada saat arus maksimum pertama

$\sin(3\pi t) = 1$

$(3\pi t) = \text{Arc sin}(1)$

$(3\pi t) = \frac{\pi}{2}$

$t = \frac{\frac{\pi}{2}}{3\pi} = \frac{1}{6}$ s $= 166,67$ ms

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c. waktu t pada saat arus i = 16 kedua

$i(t) = 32 \sin(3\pi t)$

$16 = 32 \sin(3\pi t)$

$\sin(3\pi t) = \frac{16}{22} = \frac{1}{2}$

$3\pi t = \text{Arc} \sin \frac{1}{2}$

$3\pi t = \frac{\pi}{6}$ dan $3\pi t = \frac{5\pi}{6}$

$t = \frac{\frac{\pi}{6}}{3\pi} = \frac{1}{18} s = 55,556 ms$ $t = \frac{\frac{5\pi}{6}}{3\pi} = \frac{5}{18} s = 277,78 ms$

Jadi, Arus (i) = 16 A kedua terjadi pada saat t = 277,78 ms

2. Tegangan bolak-balik suatu rangkaian listrik pada setiap saat t diberikan oleh persamaan $V = 70 \sin(100\pi t + 0,380)$. Hitung

a. Tegangan pada t = 0 dan t = 5 ms

$V(t) = 70 \sin(100\pi t + 0,380)$

t=0, $V(0) = 70 \sin(100\pi(0) + 0,380)$

$= 70 \sin(0,380)$

$= 70 \cdot 6,6322 \cdot 10^{-3} = 0,46425 V$

$t = 5 ms \rightarrow 5 \cdot 10^{-3}$ $V(5 \cdot 10^{-3}) = 70 \sin(100\pi(5 \cdot 10^{-3}) + 0,380)$

$= 70 \sin(1,95)$

$= 70 \cdot 0,034027350 5 = 2,381914535 V$

b. Waktu t pada saat tegangan mencapai maksimum

$\sin(100\pi t + 0,380) = 1$

$100\pi t + 0,380 = \text{arc} \sin 1$

$100\pi t + 0,380 = \frac{\pi}{2}$

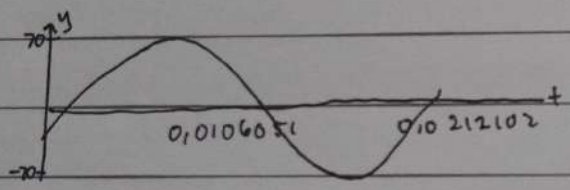
$t = \frac{\frac{\pi}{2} - 0,380}{100\pi}$

$= 3,8 \cdot 10^{-3} = 3,8 ms$

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c. Sketsa grafik osilasi tegangan untuk satu putaran



Amplitudo = 70

Waktu periodik = $\frac{2\pi}{\omega} = \frac{2\pi}{100\pi} = 0,02$ s

Frekuensi = $\frac{1}{T} = \frac{1}{0,02} = 50$ Hz

Pergeseran fase = $-\frac{a}{\omega} = -\frac{0,380}{100\pi} = -0,0021020$

(lagging)